

Special Session Proposal

Explainable AI for Smart Healthcare, Industry 5.0 and Intelligent Automation (XAI-SHIA 2027)

A brief description of the proposed area with significance and alignment to the conference:

Artificial Intelligence is rapidly transforming healthcare, industrial systems, automation, robotics, cyber-physical systems, and decision-support platforms. However, as AI models become increasingly complex, there is a growing need to ensure that their decisions are transparent, interpretable, reliable, and trustworthy. The proposed special session, “**Explainable AI for Smart Healthcare, Industry 5.0 and Intelligent Automation,**” focuses on the development and application of Explainable Artificial Intelligence techniques for intelligent systems where human trust, safety, accountability, and decision justification are essential.

Explainable AI plays a vital role in **smart healthcare**, where AI-based systems are used for disease diagnosis, medical image analysis, patient monitoring, clinical decision support, personalized treatment, and healthcare risk prediction. In such sensitive domains, black-box predictions are not sufficient. Medical professionals require understandable explanations to validate AI-based recommendations and support responsible clinical adoption.

In the context of **Industry 5.0**, AI is moving beyond automation and productivity toward human-centric, resilient, and sustainable industrial systems. Explainable AI can support predictive maintenance, industrial quality control, process optimization, smart manufacturing, robotics, human-machine collaboration, and cyber-physical production systems. By making AI decisions transparent, industries can improve operational trust, reduce risks, and enhance the collaboration between humans and intelligent machines.

For **intelligent automation**, Explainable AI is important in autonomous systems, smart control, IoT-enabled automation, edge intelligence, digital twins, robotics, and real-time decision-making. As automation systems are increasingly deployed in critical environments, explainability becomes essential for debugging, compliance, safety assurance, and user acceptance.

This special session is highly aligned with the broader theme of emerging electronics, intelligent systems, and automation, as it connects AI, healthcare technology, industrial intelligence, embedded systems, IoT, robotics, and cyber-physical automation. Unlike general AI or machine learning tracks, this session specifically emphasizes **explainability, trustworthiness, transparency, and responsible AI deployment** across smart healthcare and industrial automation applications. The session will encourage interdisciplinary research that bridges computer science, electronics, biomedical systems, industrial engineering, and automation technologies.

The session aims to provide a focused platform for researchers, academicians, industry experts, and practitioners to present novel algorithms, frameworks, tools, case studies, and real-world applications of Explainable AI in high-impact domains. It will also promote discussion on challenges such as model interpretability, ethical AI, human-AI collaboration, explainable deep learning, privacy-preserving intelligence, and AI governance in critical application areas.

Details of Session Chairs/Organizers

Organizer 1:

Name : Dr. R. Nidhya
Affiliation : Professor, Department of Computer Science and Engineering
Institution : Madanapalle Institute of Technology & Science Deemed to be University
Area of Expertise : Artificial Intelligence, Machine Learning, Data Science, IoT, Smart Systems and Healthcare AI
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Photo :

Brief Bio:

Dr. R. Nidhya is an accomplished academician and researcher in Computer Science and Engineering with over 17 years of experience in teaching, research, administration, and industry interaction. She is currently serving as Professor in the Department of Computer Science and Engineering at Madanapalle Institute of Technology & Science Deemed to be University, Andhra Pradesh. She holds Ph.D., M.Tech., and B.Tech. degrees from Anna University, Chennai, and is currently pursuing postdoctoral research at the University of Malta, Malta. Her research interests include Artificial Intelligence, Machine Learning, Wireless Body Area Networks, Security, and IoT. She has published several research papers in reputed journals and conferences and has 7 patents, 6 copyrights, and 7 books to her credit. She also serves as a reviewer for SCI/Scopus-indexed journals from reputed publishers such as Wiley and Elsevier.

Dr. Nidhya has demonstrated strong leadership in organizing funded academic and research events. She served as Coordinator for the **2nd International Conference on Optimization Techniques**, funded by AICTE with ₹3,00,000, and also coordinated a **JNTUA-funded FDP on Quantum Neuromorphic Computing for Sustainable Generative AI** with funding of ₹1,00,000. She successfully organized one IEEE International conference demonstrating her leadership in academic event management and research collaboration. She has also organized several FDPs, workshops, STTPs, induction programs, webinars, and technical events funded by AICTE, Anna University, JNTUA, CSI, and ISTE. She has mentored over 50 UG and PG projects and served as session chair, guest of honour, and resource person in numerous academic programs. She is a Senior IEEE member. She is also a life member of CSI and ISTE.

Organizer 2:

Name : Dr. M. Sreedevi
Affiliation : Professor & HoD, Department of Computer Science and Engineering
Institution : Madanapalle Institute of Technology & Science Deemed to be University
Area of Expertise : Artificial Intelligence, Machine Learning, Data Science, IoT, Smart Systems and Healthcare AI
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Brief Bio:

Dr. M. Sreedevi is a distinguished Professor in Computer Science and Engineering with strong academic, research, and administrative experience. She holds a Ph.D. in Network Security from Sri Venkateswara University, Tirupati, M.Tech. in Information Technology from Punjabi University, Patiala, and B.Tech. in Electronics and Communication Engineering from Sri Venkateswara University. Her research interests include Wireless Body Area Networks, Machine Learning, and Security. She has contributed actively to research through publications in Scopus-indexed conferences and journals, including work in healthcare systems, cloud computing, machine learning, wireless sensor networks, post-quantum cryptography, and recommendation systems.

Dr. Sreedevi has demonstrated strong leadership in organizing funded academic and research events. She served as Co-Coordinator for the **2nd International Conference on Optimization Techniques**, funded by AICTE with ₹3,00,000, and also coordinated a JNTUA-funded FDP on Quantum Neuromorphic Computing for Sustainable Generative AI with funding of ₹1,00,000. In addition, she coordinated an AICTE Prerana Workshop funded with ₹7,00,000, reflecting her capability in securing and managing academic funding. Her experience in research, funded event coordination, conference organization, and emerging technology areas makes her well suited for contributing to conference special sessions and academic research initiatives.

Organizer 3:

Name : Mr T Thangarasan
Affiliation : Assistant Professor, Department of Computer Science and Engineering
Institution : Madanapalle Institute of Technology & Science Deemed to be University
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Area of Expertise : Artificial Intelligence, Machine Learning and Healthcare AI



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Brief Bio

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Mr. Thangarasan T, is an Assistant Professor in the Department of Computer Science & Engineering with over 13 years of teaching and research experience. He is currently pursuing a Ph.D. in Computer Science and Information Engineering at Anna University. His research interests include Artificial Intelligence, Machine Learning, Deep Learning, Internet of Things (IoT), Computer Networks, Edge

Computing, and Explainable AI. He has published numerous research papers in Scopus, IEEE Xplore, ESCI, and other reputed journals, and has also filed patents in the fields of healthcare and IoT. He actively participates in faculty development programs and is committed to advancing research in intelligent computing, healthcare analytics, and emerging technologies.

Technical Outline of the Session and Topics:

1. Explainable Artificial Intelligence models for smart healthcare applications
2. Interpretable machine learning and deep learning for medical diagnosis
3. Explainable AI for medical image analysis and clinical decision support
4. Trustworthy AI for patient monitoring and personalized healthcare
5. Explainable AI for Industry 5.0 and smart manufacturing
6. Human-centric AI for industrial automation and decision support
7. Explainable AI in robotics, autonomous systems, and intelligent control
8. AI-enabled predictive maintenance with interpretable decision models
9. Explainable deep learning for cyber-physical systemsEdge
10. AI and explainability in IoT-enabled automation systems
11. Digital twin-based explainable intelligence for smart industries
12. Federated learning and privacy-preserving explainable AI
13. Explainable AI for quality inspection and industrial fault diagnosis
14. Ethical, transparent, and responsible AI for critical applications
15. Human-AI collaboration and decision transparency in automation
16. Explainable reinforcement learning for intelligent control systems
17. AI governance, fairness, bias detection, and accountability in smart systems
18. Case studies and real-time applications of explainable AI in healthcare and industry

Potential Contributing Authors

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4. Dr. A Dineshkumar, KL Deemed to be University, dineshngpit@gmail.com
5. Dr. D Radha, Arifa Institute of Technology, dradha2009@gmail.com
6. Research Scholars and faculty members working in AI, healthcare analytics, Industry 5.0, automation, IoT, cyber-physical systems, and intelligent computing may be invited to contribute papers to this session.

Potential Reviewers

Names, affiliations and emails, at least 20:

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Other Details:

This special session will invite original research articles, review papers, case studies, and application-oriented contributions related to Explainable AI in healthcare, Industry 5.0, and intelligent automation. The session is expected to attract contributions from researchers working in Artificial Intelligence, Machine Learning, Deep Learning, Biomedical Computing, Smart Manufacturing, Robotics, IoT, Edge Computing, Cyber-Physical Systems, and Automation.

The session will focus on both theoretical and applied aspects of Explainable AI, including algorithmic development, model interpretation, real-time implementation, ethical AI, and deployment challenges. It will provide a platform for academic and industry researchers to discuss how explainable and trustworthy AI can improve safety, transparency, reliability, and adoption of intelligent systems in critical domains.

The proposed session will be suitable for interdisciplinary participation and will strengthen the conference by adding a focused theme on responsible, transparent, and application-driven AI for next-generation healthcare and automation systems.